

A mean-field theory for the blackboard game

Derivation, solution, and an honest comparison with the exact simulation

MA-CC blackboard-control study

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Abstract

We derive a mean-field theory for the blackboard game from its microscopic rules, with **no fitted parameter**: every coefficient is a game setting (ρ, N, q, b) . We solve it numerically and compare it with the exact simulation. **It does not match**: mean absolute error 0.25 in the population coordinate, and in the controlled arms the theory gets the *direction* wrong. We identify the cause precisely — a product-form closure cannot represent a posterior whose value depends on whether particular facts *co-occur* — and state what the theory needs. We also give the continuous-time limit and the Ornstein–Uhlenbeck reduction that the closure would permit if it worked.

1 What to track, and why

The game’s only epistemic dynamics are *decay* and *exposure-activation*, and both act on an (agent, fact) pair. So the natural mean-field variable is the occupancy

$$m_{if}(t) = \mathbb{P}(\text{fact } f \text{ is active for agent } i \text{ at round } t).$$

The population coordinate we measure follows by averaging the exact posterior,

$$M(t) = \frac{1}{N} \sum_i \mathbb{E}_{K_i} [\mathbb{P}(\text{ALLOCATION_0} \mid K_i)], \quad K_i \sim \prod_f \text{Bernoulli}(m_{if}), \quad (1)$$

and Eq. (1) is the **closure**: it asserts that an agent’s facts are independent given their marginals. Everything that follows is exact except this.

2 Derivation, one round

In the order the runtime applies them.

1. Decay. Each active fact survives with probability ρ :

$$m_{if} \longmapsto \rho m_{if}. \quad (2)$$

2. The board. Agent i posts one fact it holds. Writing π_{if} for the probability it chooses f , and approximating the choice as uniform over held facts, $\pi_{if} = m_{if} / \sum_g m_{ig}$. The controller adds its b facts, indicated by $c_f \in \{0, 1\}$. The per-post marginal is $\bar{\pi}_f = \frac{1}{N} \sum_i \pi_{if}$.

3. Exposure. This is where the microscopic detail matters. Agents are updated *sequentially* and the board *builds up within the round*, so the agent in position k reads only the k posts made before it. With per-post probability $\bar{\pi}_f$, the chance that position k sees f is $1 - (1 - \bar{\pi}_f)^k$, and averaging over positions gives a closed form:

$$\bar{u}_f = \frac{1}{N} \sum_{k=0}^{N-1} \left[1 - (1 - \bar{\pi}_f)^k \right] = 1 - \frac{1 - (1 - \bar{\pi}_f)^N}{N \bar{\pi}_f}. \quad (3)$$

A *synchronous* mean field would instead use $u_f = \mathbf{1}\{B_f \geq 1\}$, which with full board reading sends every occupancy to one after a single round. Equation (3) is the damping that replaces it. A controller post is on the board before any agent reads, so it is seen by everyone: $u_f = 1 - (1 - \bar{u}_f)(1 - c_f)$.

Under **uniform** sampling of q of n messages, the hypergeometric form replaces Eq. (3), $u_f = 1 - \binom{n-k_f}{q} / \binom{n}{q}$, which is exactly Eq. (7) of the task_004 paper.

4. Combine. A fact is active afterwards if it survived decay or was (re)activated:

$$m_{if}(t+1) = \rho m_{if}(t) + (1 - \rho m_{if}(t)) u_f(t) \quad (4)$$

What it depends on. Only ρ , N , q , b , the controller rule, and the initial assignment. **Nothing is fitted.** That is the theory’s strongest claim and it is what makes the comparison below a real test.

3 Continuous time, and the Ornstein–Uhlenbeck reduction

Writing $\rho = e^{-\gamma}$ and taking one round as dt , Eq. (4) becomes

$$\dot{m}_{if} = -\gamma m_{if} + (1 - m_{if}) \alpha_f, \quad \gamma = -\ln \rho, \quad \alpha_f = \text{exposure rate}, \quad (5)$$

a logistic relaxation with fixed point $m_f^* = \alpha_f / (\gamma + \alpha_f)$ and rate $\gamma + \alpha_f$. This is the discrete-to-continuous correspondence: the **deactivation rate** is $-\ln \rho$, exactly the γ of the task_004 paper’s Table 1, and the per-round map is its Euler step. The two descriptions agree to $O(dt^2)$ and diverge when γdt is not small, i.e. at low ρ : at $\rho = 0.6$, $\gamma = 0.51$ per round, so a one-round step is already a 50% correction and the discrete map should be preferred.

If the closure held, linearising Eq. (5) about m^* and adding finite- N sampling noise would give the three-variable Ornstein–Uhlenbeck system requested earlier,

$$dM = -\theta(M - M_\infty)dt + g_p S^p dt + g_c S^c dt + \sigma_M dW, \quad dS^p = -\theta_p(S^p - M)dt + \sigma_p dW_p,$$

$$dS^c = -\theta_c(S^c - u \pi(M))dt + \sigma_c dW_c,$$

with $\theta = -\ln \rho$, $M_\infty = 1/3$, the peer force reverting to M itself, and the controller force reverting to the activation policy evaluated at M — the feedback. Its stationary covariance would follow from the Lyapunov equation and the path KL in closed form from Girsanov. **We do not report those numbers**, because the closure fails first.

4 The comparison, and it fails

setup	arm	sim e (r29)	theory M	error
nosolution-v2 $\rho=1$	silent	0.779	0.500	−0.279
nosolution-v2 $\rho=1$	false	0.125	0.500	+0.375
nosolution-v2 $\rho=0.75$	silent	0.545	0.504	−0.041
symmetric-v2 $\rho=0.75$	silent	0.598	0.759	+0.161
symmetric-v2 $\rho=0.75$	false	0.067	0.654	+0.587

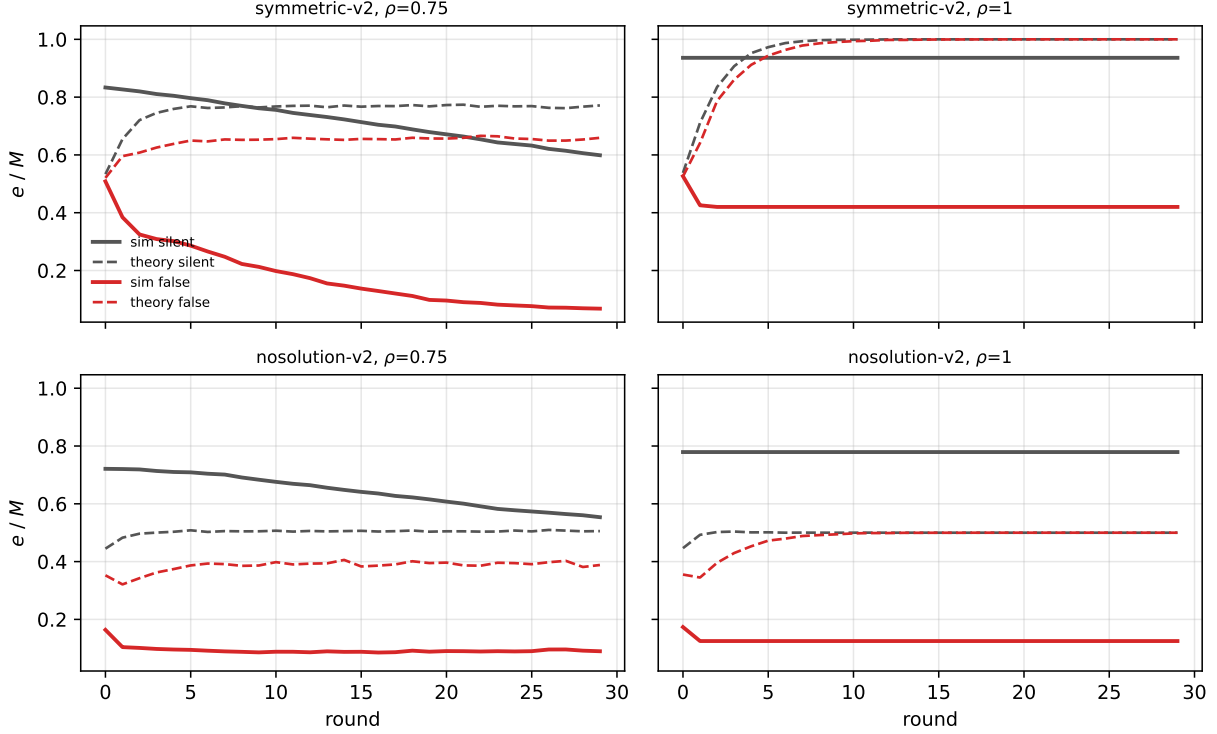


Figure 1: Theory (dashed) against the exact simulation (solid), no fitting. Silent arms in grey, false-control arms in red.

Mean absolute error 0.252, maximum 0.587. The silent arms are sometimes close (symmetric-v2 at $\rho = 0.75$ agrees to 0.03 over rounds 2–10); the **controlled arms are wrong in direction**, the theory rising where the simulation falls.

5 Why it fails, precisely

The closure, not the dynamics. The occupancy map (4) is close to exact. The failure is Eq. (1): the posterior $\mathbb{P}(\text{ALLOCATION_0} \mid K)$ is a strongly nonlinear function of *which* facts co-occur. In symmetric-v2 the six decisive facts prove **ALLOCATION_0** only *together*; a product measure with the right marginals almost never realises that conjunction, and almost never realises the conjunctions that make **ALLOCATION_2** favourite either. Averaging over independent facts therefore pulls every agent toward a generic mid-range posterior — which is exactly the $M \approx 0.5$ –0.76 plateau the theory shows.

Consequence for the controlled arms. The simulated oracle drives e to 0.067 by placing *a few specific* **ALLOCATION_2**-favouring facts in everyone’s hands. Under a product measure those facts are diluted among the agent’s other facts and their joint effect never forms. The theory consequently sees the controller as adding evidence in general, which raises M .

What to change. In increasing order of cost:

1. **Keep the exact conjunction structure for a small subset.** Track the joint law of the 6 decisive facts (64 states) exactly and the rest by marginals. Cheap, and it is where the nonlinearity lives.
2. **Pair correlations.** Add m_{ifg} for pairs, closing at third order. $\sim 20^2/2$ numbers per agent; still trivial to integrate.

3. **Change the coarse variable.** Track not $\mathbb{E}[\mathbb{P}(\text{ALLOCATION_0}|K)]$ but the distribution of *proof coverage* — how many of each agent’s minimal proofs are complete. That variable is closer to additive in the facts.

Does the theory need fitting? No, and it should not be fitted. Every coefficient is a game setting. If a fitted parameter were introduced to close the 0.25 gap it would absorb the closure error and the theory would stop being a test of anything. The honest statement is that the product-form mean field is *falsified* for this observable, by a simulation that costs 28 seconds to run.

6 What this exercise established

The mean-field programme is not dead; its first closure is. The occupancy dynamics (4), the sequential-exposure correction (3) and the identification $\gamma = -\ln \rho$ are all reusable. What fails is the assumption that an agent’s facts are independent — and that assumption is exactly what the game is designed to violate, since the whole point of a *decisive set* is that certain facts matter only in conjunction.

That is a result worth having before any of it reaches a paper, and it was obtained for free.